Example 1: Can a triangle be formed with 7 cm, 10 cm and 5 cm as its sides?

Solution: Instead of checking triangle inequality by all the sides in the triangle, check only with two smaller sides.

Sum of two smaller sides of the triangle = 5+7=12 cm > 10 cm, the third side.

It is greater than the third side.

So, a triangle can be formed with the given sides.

Example 2: Can a triangle be formed with 7 cm, 7 cm and 7 cm as its sides?

Solution: If three sides are equal, then definitely a triangle can be formed, as the triangle inequality is satisfied.

Example 3: Can a triangle be formed with 8 cm, 3 cm and 4 cm as its sides?

Solution: The sum of two smaller sides = 3+4=7 cm <8 cm, the third side.

It is less than the third side.

So, a triangle cannot be formed with the given sides.

Try these

Can a triangle be formed with the given sides? If yes, state the type of triangle formed.

S.No.	$\overline{AB}$	$\overline{BC}$	$\overline{CA}$	Can a ΔABC be formed?	Type of triangle
1	7 cm	10 cm	6 cm		
2	10 cm	8 cm	8 cm		
3	8.5 m	7.3 m	6.8 m		
4	4 cm	5 cm	12 cm		
5	15 m	20 m	20 m		
6	23 cm	20 cm	18 cm		
7	3.2 cm	1.5 cm	1.5 cm		

Example 4: Can a triangle be formed with the angles 80°, 30°, 40°?

Solution: The sum of three angles = 80°+ 30°+ 40° = 150° **(not equal to 180°)**

In a triangle, the sum of three angles is 180°.

So, a triangle cannot be formed with the given angles.

Think

Can the difference between two larger sides be less than the third side?

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